

PharmaChain

AI/ML Project Requirements Document (PRD)

Smart India Hackathon 2026

AI/ML Architecture for Pharmaceutical Supply-Chain Integrity, Anomaly Detection and
Physical Packaging Verification

Document Scope: Traditional Machine Learning + Computer Vision

Deployment: FastAPI + Docker

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1. Executive Summary

PharmaChain is an AI/ML-enabled pharmaceutical supply-chain integrity system designed to identify suspicious activity across the movement and handling of medicines from manufacturers through wholesalers and retailers to customers. The AI/ML layer is divided into two complementary capabilities: **Traditional Machine Learning** for behavioral and transactional anomaly detection, and **Computer Vision** for physical packaging and label verification.

The Traditional ML layer analyzes structured supply-chain records stored in MongoDB. It uses unsupervised anomaly-detection techniques, primarily **DBSCAN** and **Isolation Forest**, to identify unusual movement, timing, inventory, and batch-level behavior without requiring a large pre-labeled fraud dataset. Detected events are converted into four operational risk levels: **Normal, Medium, High, and Critical**.

The Computer Vision layer is planned as the physical-verification component. It will use **YOLO** for object/region detection and **OCR** with EasyOCR for extracting and comparing medicine names, symbols, labels, and other packaging elements. This layer is intended to detect subtle tampering or counterfeit modifications that may be difficult to notice visually.

Both AI/ML capabilities will be exposed through **FastAPI** endpoints and packaged into a **Docker** image, allowing the intelligence layer to integrate cleanly with the broader PharmaChain application architecture.

2. Problem Statement

Pharmaceutical supply chains can be compromised by counterfeit products, manipulated packaging, unauthorized diversion, suspicious logistics patterns, inventory hoarding, and other forms of supply-chain abuse. Conventional rule-based checks may identify obvious violations but can struggle to detect subtle or previously unseen patterns.

PharmaChain addresses this gap by combining structured-data anomaly detection with physical package inspection. The objective is not merely to determine whether a single transaction is valid, but to establish whether the observed behavior is consistent with the expected pharmaceutical supply-chain journey and whether the physical medicine package appears authentic.

3. Product Scope

Area	Requirement
User views	Support the manufacturer, wholesaler, retailer, and customer views within the overall PharmaChain system.
Data intelligence	Analyze manufacturer, wholesaler, and retailer events stored in MongoDB.
Anomaly detection	Detect unusual temporal, route, batch, inventory, and identity-related patterns.
Risk scoring	Classify detected events into Normal, Medium, High, and Critical risk levels.
Physical verification	Provide a planned CV layer for package/label tampering and subtle text/symbol changes.
API integration	Expose AI/ML inference through FastAPI endpoints.
Deployment	Containerize the AI/ML service using Docker.

4. User-View and AI/ML Interaction

Actor	Relevant Data / Interaction	AI/ML Role
Manufacturer	Medicine name, production date, export date, batch ID and related production/shipping information.	Establish the expected origin and temporal baseline for a batch; contribute features for anomaly detection.
Wholesaler	Import date, batch ID, sell date and related movement information.	Detect abnormal arrival/sale timing, route inconsistencies and suspicious batch movement.
Retailer	Receipt/stock movement information, batch ID, sale-related information and inventory behavior.	Detect unusual inventory patterns, withholding, hoarding and downstream anomalies.
Customer	Medicine/QR/package information presented for verification.	Support authenticity checks using supply-chain intelligence and, in the planned CV layer, package/label inspection.

5. AI/ML Architecture Overview

Primary pipeline: MongoDB → feature preparation → anomaly models → anomaly aggregation/risk classification → FastAPI → PharmaChain application.

Planned physical-verification pipeline: Medicine/package image → YOLO detection → relevant crop/region extraction → EasyOCR text extraction → comparison/consistency checks → tampering/counterfeit signal → FastAPI.

The two layers are complementary. Traditional ML evaluates whether the *digital supply-chain history* of a medicine is plausible, while Computer Vision evaluates whether the *physical package presentation* is consistent with expected medicine information.

6. Traditional Machine Learning Requirements

The Traditional ML subsystem is an unsupervised anomaly-detection engine. It is designed to work primarily with structured supply-chain events and does not depend on a large manually labeled dataset of fraudulent transactions.

The data source is a **MongoDB** database containing information updated by manufacturers and downstream participants. The ML pipeline converts these records into numerical and temporal features suitable for anomaly detection.

6.1 Data Requirements

Source	Representative fields	Purpose
Manufacturer	Medicine name, production date, export date, batch ID	Establish batch origin, production/export timeline and identity.
Wholesaler	Import date, batch ID, sell date	Track downstream movement and timing.
Retailer	Batch ID, sale/inventory events and related records	Analyze final distribution and inventory behavior.

6.2 Feature Engineering

- **Temporal features:** production-to-export duration, export-to-import duration, import-to-sale duration, total transit/holding duration.
- **Batch features:** batch frequency, duplicate/parallel appearances, unexpected reuse, sequence inconsistencies.
- **Route/movement features:** movement sequence, geographic/route consistency where location data is available, repeated or impossible transitions.
- **Inventory features:** stock accumulation, unusually long holding periods, abnormal purchase/sale velocity, sudden quantity concentration where quantity data is available.
- **Identity/QR features:** repeated scans or repeated batch/QR appearances across incompatible locations or timelines.

6.3 Models

Model	Role in PharmaChain	Expected signal
DBSCAN	Density-based clustering of supply-chain behavior and identification of points that do not belong to normal behavioral clusters.	Sparse or isolated behavioral patterns, unusual route/timing combinations, outlying transaction groups.
Isolation Forest	Tree-based unsupervised anomaly detection that isolates observations requiring fewer random splits than typical observations.	Individual transactions/batches exhibiting unusual feature combinations.

Using both approaches provides complementary anomaly signals: DBSCAN captures density/cluster structure, while Isolation Forest provides an independent observation-level anomaly score.

7. Anomaly Scenarios

Scenario	AI/ML interpretation
Impossible travel	The same batch/medicine appears at locations or supply-chain points in a sequence that is implausible within the observed timestamps and route constraints.
Batch withholding	A batch remains at a participant or stage substantially longer than the normal behavioral pattern.
Too-late arrival	Observed import/arrival timing is an extreme outlier relative to expected movement duration.
Cloned QR	A QR/batch identity appears in multiple incompatible transactions, locations, or time windows, suggesting duplication or reuse.
Route infiltration	A batch exhibits an unexpected participant/route transition inconsistent with its normal supply-chain path.
Inventory hoarding	Inventory remains unusually concentrated or stagnant compared with expected downstream movement.

8. Risk Classification Requirements

Model outputs should be transformed into an operational risk classification. The initial design uses four levels. Exact numerical thresholds should be calibrated during testing against observed data and domain expectations rather than hard-coded prematurely.

Risk	Meaning	Example action
Normal	No significant anomaly signal.	Allow normal processing; retain audit trail.
Medium	Weak or localized anomaly requiring attention.	Flag for monitoring or secondary verification.
High	Strong anomaly signal or multiple suspicious indicators.	Escalate for investigation and additional verification.
Critical	Severe/multiple anomalies indicating a potentially compromised batch or counterfeit event.	Immediate escalation, hold/verification workflow, and investigation.

9. Planned Computer Vision Layer

The Computer Vision layer is currently a **draft architecture** and has not yet been implemented. Its purpose is to detect tampering in the physical packaging of medicine units, including subtle text and symbol modifications that can be difficult for a human to notice.

For example, if an authentic medicine name is **CALPOL** but a counterfeit package changes it to a visually similar misspelling such as **COLPAL**, the CV/OCR pipeline should identify the discrepancy against the expected medicine identity. The same principle applies to added, removed, altered, or substituted symbols and packaging elements.

9.1 Proposed CV Pipeline

- **Image acquisition:** receive a photograph or camera frame of the medicine package.
- **YOLO detection:** identify the relevant package, label, medicine-name region, logo/symbol region, QR/barcode region, or other trained/defined visual targets.
- **Region extraction:** crop the relevant detected areas to improve OCR quality and reduce irrelevant visual content.
- **OCR:** use EasyOCR to extract visible text from the selected region.
- **Consistency comparison:** compare extracted text and expected medicine metadata, including medicine name and potentially other identifiers.
- **Tampering signal:** identify suspicious spelling differences, missing/extra symbols, altered labels, or other defined visual inconsistencies.
- **API response:** return structured verification results to the PharmaChain application.

9.2 CV Technology Stack

Technology	Planned responsibility
YOLO	Object/region detection for medicine packaging and relevant visual components.
OpenCV	Image preprocessing, cropping, resizing, normalization and supporting computer-vision operations.
EasyOCR	OCR-based extraction of medicine names and other visible textual information.

10. System Integration & Deployment

The AI/ML subsystem will be deployed as an API-oriented service so that the main PharmaChain application can invoke anomaly detection and, once implemented, physical package verification without embedding ML logic directly into the frontend or business-logic layer.

Component	Requirement
MongoDB	Persist manufacturer, wholesaler, retailer and related supply-chain records used by the ML pipeline.
Python ML service	Implement data preparation, feature engineering, DBSCAN, Isolation Forest, anomaly aggregation and risk classification.
FastAPI	Expose inference/verification endpoints for integration with the PharmaChain application.
Docker	Package the service and its dependencies into a reproducible deployment image.

Indicative endpoint design: the final API contract should be finalized with the application team. A possible separation is an anomaly-analysis endpoint for structured batch/transaction data and a package-verification endpoint for image input.

11. Technology Stack

Layer	Technology	Purpose
Data processing	Pandas	Data loading, transformation, feature preparation and tabular processing.
Numerical computing	NumPy	Numerical operations and model input preparation.
Machine Learning	Scikit-learn	DBSCAN, Isolation Forest, preprocessing and supporting ML utilities.
Database	MongoDB	Supply-chain records and manufacturer/wholesaler/retailer event data.
Computer Vision	YOLO	Planned packaging/object/region detection.
Image processing	OpenCV	Planned image preprocessing and vision operations.
OCR	EasyOCR	Planned text extraction from medicine packaging.
API	FastAPI	Model inference and application integration endpoints.
Deployment	Docker	Containerized deployment of the AI/ML service.

12. Functional Requirements

ID	Requirement
FR-01	The system shall ingest or query relevant manufacturer, wholesaler and retailer supply-chain records from MongoDB.
FR-02	The system shall construct anomaly-detection features from timestamps, batch identifiers, movement information and available inventory/transaction data.
FR-03	The system shall support DBSCAN-based unsupervised behavioral analysis.
FR-04	The system shall support Isolation Forest-based anomaly scoring.

ID	Requirement
FR-05	The system shall combine model outputs and relevant rule/context signals into an operational anomaly assessment.
FR-06	The system shall classify anomaly events into Normal, Medium, High and Critical risk levels.
FR-07	The system shall return an explainable response containing the detected anomaly indicators contributing to the alert, where feasible.
FR-08	The system shall expose the Traditional ML functionality through FastAPI.
FR-09	The planned CV service shall accept medicine/package images and identify relevant visual regions using YOLO.
FR-10	The planned CV service shall extract relevant packaging text using EasyOCR.
FR-11	The planned CV service shall compare extracted information against expected medicine information and generate a tampering/counterfeit signal.
FR-12	The AI/ML service shall be containerizable and deployable using Docker.

13. Non-Functional Requirements

- **Performance:** inference should be sufficiently fast for the intended application workflow; latency targets should be finalized after endpoint and hardware selection.
- **Reliability:** invalid, missing, or incomplete records/images should produce controlled API responses rather than service failure.
- **Scalability:** the service should be stateless where practical so multiple Docker instances can be deployed if system traffic increases.
- **Explainability:** anomaly responses should expose relevant indicators such as abnormal duration, duplicate identity, route inconsistency, or inventory deviation.
- **Reproducibility:** Python dependencies, model versions and configuration should be pinned/documented in the Docker build.
- **Security:** API access, database credentials, uploaded images and sensitive supply-chain information should be protected through appropriate authentication, secrets management and access controls.
- **Auditability:** alerts should preserve enough metadata to trace the underlying batch/event and support human investigation.

14. Model Development & Validation Requirements

Because the initial ML approach is unsupervised, model quality should not be evaluated solely using conventional classification accuracy. Validation should focus on whether the models identify meaningful deviations and whether alerts are useful to domain users.

Area	Requirement
Data quality	Validate missing values, duplicate records, inconsistent timestamps and invalid batch identifiers before model inference.
Feature validation	Check that engineered features reflect actual supply-chain behavior and do not introduce leakage.

Area	Requirement
Threshold calibration	Tune anomaly and risk thresholds using historical/controlled scenarios and expert review.
Scenario testing	Create test cases representing impossible travel, withholding, late arrival, cloned QR, route infiltration and hoarding.
Human review	Have suspicious outputs reviewed against expected operational behavior before production decisions are automated.
CV validation	For the planned CV layer, measure detection and OCR performance on authentic and intentionally tampered packages.

15. Example AI/ML Output

A recommended response structure for the application is a machine-readable object containing the batch/transaction identity, anomaly score(s), triggered indicators, risk level, and supporting metadata. The exact schema should be aligned with the broader PharmaChain backend.

Illustrative response: Batch ID → PH-2026-00192; anomaly signals → unusually long holding duration + incompatible movement timing; Isolation Forest → anomalous; DBSCAN → outlier; risk → High; recommended action → secondary verification/investigation.

For CV, an illustrative response could contain: medicine expected → CALPOL; OCR result → COLPAL; text mismatch → true; symbol anomaly → false; packaging detection confidence → available; tampering risk → High.

16. Risks, Limitations & Mitigations

Risk / Limitation	Impact	Mitigation
Limited labeled fraud data	Harder to train/evaluate supervised fraud models.	Use unsupervised detection initially and build a reviewed alert dataset over time.
Poor data quality	Can create false anomalies.	Validate and normalize records before feature generation.
False positives	Legitimate unusual events may be flagged.	Use multi-signal scoring, calibrated thresholds and human review for high-risk cases.
Missing route/location data	Impossible-travel detection may be limited.	Use route features only where trustworthy location/time data exists; otherwise rely on temporal/batch signals.
OCR errors	Image quality or fonts may cause incorrect text extraction.	Use preprocessing, confidence thresholds and comparison rules; validate on representative packaging images.
Subtle visual changes	Counterfeits may evade basic detection.	Expand CV training/evaluation to real and controlled tampering examples.

17. Implementation Roadmap

Phase	Deliverables	Status
Phase 1	MongoDB schema/data integration; data cleaning and feature engineering for supply-chain events.	Core scope
Phase 2	DBSCAN + Isolation Forest pipeline; anomaly scoring and initial risk classification.	Core ML scope
Phase 3	Scenario-based validation for impossible travel, withholding, late arrival, cloned QR, route infiltration and hoarding.	Validation stage
Phase 4	FastAPI endpoints, structured responses, logging and integration with the PharmaChain application.	Planned
Phase 5	Docker packaging and deployment testing.	Planned
Phase 6	YOLO + OpenCV + EasyOCR CV implementation; packaging dataset creation and tampering tests.	Draft; not implemented
Phase 7	End-to-end integration of digital anomaly intelligence + physical package verification.	Future integration

18. Acceptance Criteria

- A representative supply-chain dataset can be read from MongoDB and transformed into model-ready features without manual intervention.
- DBSCAN and Isolation Forest execute successfully on the prepared feature set and produce anomaly indicators/scores.
- The system can flag representative examples of the defined anomaly scenarios when their data signatures are present.
- Risk outputs are mapped consistently to Normal, Medium, High and Critical categories using documented thresholds/logic.
- The FastAPI service returns a stable, documented JSON response for valid and invalid requests.
- The AI/ML service can be built and run as a Docker image.
- The planned CV architecture can be implemented as an independent endpoint without changing the core Traditional ML pipeline.
- The CV prototype, once implemented, demonstrates detection of representative medicine-name alterations and selected symbol/packaging modifications on a controlled test set.
- High/Critical alerts retain sufficient evidence or metadata for a human reviewer to understand why the alert was generated.

19. End-to-End Conceptual Architecture

Supply-chain intelligence path: Manufacturer / Wholesaler / Retailer data → MongoDB → Pandas/NumPy preprocessing → feature engineering → DBSCAN + Isolation Forest → anomaly aggregation → risk classification → FastAPI → PharmaChain application.

Physical verification path (planned): Customer/inspector image → YOLO → package/label region → OpenCV preprocessing → EasyOCR → expected-vs-observed comparison → tampering signal → FastAPI → PharmaChain application.

Decision layer: The broader PharmaChain platform can use both digital supply-chain anomalies and physical package inconsistencies as independent or combined evidence for verification and investigation.

20. Future Enhancements

- Introduce a feedback loop in which investigator decisions label historical alerts, enabling future supervised or semi-supervised fraud models.
- Add graph-based supply-chain analysis to model entities and relationships among manufacturers, batches, wholesalers, retailers and routes.
- Add temporal sequence models when sufficient longitudinal data becomes available.
- Introduce stronger QR/identifier integrity checks and cryptographic verification where supported by the platform.
- Expand CV from OCR-based mismatch detection toward logo, seal, font, layout and packaging-defect analysis.
- Fuse digital and visual signals into a unified confidence/risk score while preserving the individual evidence sources.

21. Summary

The PharmaChain AI/ML architecture is designed as a two-layer intelligence system. The **Traditional ML layer** uses MongoDB-backed supply-chain data and unsupervised methods such as DBSCAN and Isolation Forest to identify behavioral and transactional anomalies, including impossible travel, batch withholding, late arrivals, cloned QR activity, route infiltration and inventory hoarding. These signals are converted into four risk categories for operational action.

The **Computer Vision layer**, currently at the draft stage, will extend verification to the physical medicine package using YOLO, OpenCV and EasyOCR. Its goal is to catch subtle counterfeit/tampering indicators such as altered medicine names or symbols that can be overlooked by manual inspection.

Together, these capabilities provide PharmaChain with a practical AI/ML foundation for pharmaceutical supply-chain integrity: **digital behavior verification + physical package verification**, delivered through FastAPI and packaged with Docker for integration and deployment.